

Secure Communication Protocol for Unmanned Aerial Vehicles (UAVs)

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Abstract

The increasing deployment of Unmanned Aerial Vehicles (UAVs) in both civilian and military applications necessitate robust security mechanisms for their communication links. The widely adopted MAVLink protocol, while lightweight and efficient, lacks built-in encryption and authentication, rendering UAV systems vulnerable to eavesdropping, spoofing, message manipulation, and replay attacks. This project proposes and implements a secure communication protocol for UAVs by integrating a lightweight security layer based on the ChaCha20-Poly1305 authenticated encryption algorithm into a MAVLink-like protocol. The proposed system secures bidirectional communication between the Ground Control Station (GCS) and UAVs, including command-and-control transmissions as well as telemetry and status reporting exchanges. The system is designed to ensure confidentiality, integrity, authentication, and replay protection in an ad-hoc wireless environment. Using the ns-3 network simulator and the OpenSSL library, the proposed solution incorporates authenticated encryption and sequence number tracking mechanisms to mitigate replay attacks and unauthorized message injection. The performance of the system is evaluated in terms of packet loss, latency, throughput, cryptographic processing overhead, and energy consumption. Simulation results demonstrate that the proposed security layer effectively mitigates malicious activities, including replay and jamming attacks, while maintaining lightweight computational overhead and preserving communication efficiency. Furthermore, the proposed approach introduces only limited additional energy consumption compared with the unsecured protocol, indicating its suitability for real-time UAV operations and resource-constrained aerial platforms. Overall, the proposed framework significantly enhances the security and reliability of MAVLink-based UAV communications without compromising operational performance or real-time responsiveness.

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List of Abbreviations

Abbreviation	Definition
UAV	Unmanned Aerial Vehicle / Vehicles
GCS	Ground Control Station
MAVLink	Micro Air Vehicle Link
UDP	User Datagram Protocol
AEAD	Authenticated Encryption with Associated Data
CRC	Cyclic Redundancy Check
ACK	Acknowledgment
SEQ	Sequence Number
IP	Internet Protocol
MAC	Medium Access Control
OSI	Open Systems Interconnection
Wi-Fi	Wireless Fidelity
CSMA/CA	Carrier Sense Multiple Access with Collision Avoidance
IPv4	Internet Protocol Version 4
RF	Radio Frequency
PX4	PX4 Autopilot Platform
IoUAV	Internet of UAVs
BVLOS	Beyond Visual Line of Sight

AES	Advanced Encryption Standard
AES-GCM	Advanced Encryption Standard – Galois/Counter Mode
TLS	Transport Layer Security
ECC	Elliptic Curve Cryptography
DoS	Denial of Service
RAM	Random Access Memory
GPS	Global Positioning System
ns-3	Network Simulator 3
NetAnim	Network Animator

Introduction

1.1 Introduction

Unmanned Aerial Vehicles (UAVs), commonly known as drones, have gained significant attention in recent years due to their wide range of applications in both civilian and military domains. These applications include surveillance, aerial photography, disaster management, environmental monitoring, and logistics services. UAV systems rely on wireless communication to exchange data between the UAV and the Ground Control Station (GCS).

One of the commonly used communication protocols in UAV systems is MAVLink, which enables efficient data exchange between system components.

Despite its advantages, MAVLink was originally designed with limited security mechanisms, which may expose UAV communications to various cyber threats such as eavesdropping, spoofing, and message manipulation. Therefore, enhancing the security of UAV communication has become an important research area to ensure reliable and secure operation of these systems.

1.2 Problem Statement

UAV communication systems rely heavily on wireless networks, which makes them vulnerable to several security threats. The MAVLink protocol, widely used in UAV systems, lacks strong built-in encryption and authentication mechanisms in its basic design. This limitation may allow attackers to intercept transmitted data, inject malicious messages, or gain unauthorized control over the UAV.

As a result, ensuring secure communication between the UAV and the Ground Control Station is a critical challenge that needs to be addressed.

1.3 Project Objectives

The main objective of this project is to enhance the security of UAV communication by improving the protection mechanisms used in MAVLink-based communication systems.

The specific objectives of the project include:

- Analyzing the security vulnerabilities of UAV communication protocols.
- Studying existing security mechanisms used in UAV networks.

- Proposing an approach to enhance the security of MAVLink communication.
- Evaluating the performance and effectiveness of the proposed security solution.

1.4 Significance of the Study

The importance of this project lies in improving the security of UAV communication systems, which are increasingly used in critical applications. Enhancing communication security helps protect sensitive data, prevents unauthorized access to UAV systems, and reduces the risks associated with cyberattacks. Furthermore, this study contributes to the development of more reliable and secure UAV communication frameworks that can be applied in various real-world applications.

Chapter 1: Theoretical Concepts

1.1 Introduction

Unmanned Aerial Vehicle (UAV) systems are widely recognized as cyber-physical systems that rely on wireless communication to enable coordination between aerial platforms and ground control stations. The inherent reliance on wireless channels introduces fundamental security vulnerabilities due to the broadcast nature of radio communication, which allows adversaries to intercept, modify, or replay transmitted data. Consequently, UAV communication security has become a critical research domain, particularly with the increasing deployment of UAVs in civilian, industrial, and defense applications. Ensuring secure communication is therefore essential to maintain system reliability, mission integrity, and operational safety.

1.2 UAV System Architecture

A UAV system is typically modeled as a distributed architecture composed of an aerial platform, a ground control station, and a bidirectional wireless communication channel. The aerial platform integrates embedded flight controllers, navigation modules, and sensor systems that enable autonomous or semi-autonomous operation. The ground control station serves as the central entity responsible for mission planning, monitoring, and control command transmission. The interaction between these components forms a closed-loop control system operating over an insecure wireless medium. Due to the open nature of radio frequency channels, UAV systems are inherently exposed to external interception and manipulation. Studies have demonstrated that attackers within communication range can exploit these vulnerabilities to disrupt system behavior or extract sensitive operational data, particularly in unprotected communication environments [5], [6].

1.3 MAVLink Communication Protocol

MAVLink (Micro Air Vehicle Link) is a lightweight communication protocol designed for UAV systems and embedded aerospace applications. The protocol is optimized for low bandwidth consumption and real-time telemetry exchange, making it suitable for resource-constrained communication environments. MAVLink defines a structured binary message format consisting of system identifiers, component identifiers, message identifiers, payload data, and checksum

fields. This structure minimizes communication overhead while supporting efficient packet transmission between UAV nodes and ground control stations [13], [14].

However, MAVLink was primarily designed for communication efficiency and interoperability rather than security. Consequently, the protocol lacks built-in mechanisms for encryption, authentication, and integrity protection, exposing MAVLink-based systems to multiple cybersecurity threats [1],[4],[14].

Previous studies have demonstrated that attackers can exploit these vulnerabilities to perform message injection, replay attacks, packet manipulation, and unauthorized command transmission in unsecured UAV communication environments. Such attacks may lead to mission disruption, UAV behavior manipulation, or unauthorized control access [1], [4].

1.3.1 Packet Structure

Each transmitted packet follows a structured format inspired by the MAVLink protocol to ensure efficient parsing and communication reliability.

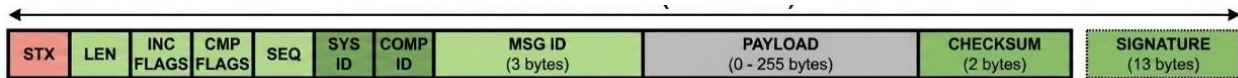


Figure1 : MAVLink Packet Structure [14]

The packet structure is defined as follows:

Table 1: MAVLink Frame Structure

Field	Size	Function
STX	1 byte	Start of frame (0xFD)
LEN	1 byte	Payload length
Incompat Flags	1 byte	Non-compatible feature flags
Compat Flags	1 byte	Compatible feature flags
Seq	1 byte	Sequence number
SysID	1 byte	System ID
CompID	1 byte	Component ID
MsgID	3 bytes	Message ID
Payload	Variable	Actual data
Checksum	2 bytes	Error detection (CRC)
Signature	13 bytes (optional)	Message authentication

This structure ensures efficient transmission, error detection, and extensibility for future enhancements.

1.3.2 HEARTBEAT Message

The HEARTBEAT message is one of the fundamental message types in the MAVLink communication protocol. It is periodically transmitted between UAV nodes and the Ground Control Station (GCS) to indicate node availability, communication status, and operational activity [13], [14]. The HEARTBEAT message enables communicating entities to detect active UAV systems, monitor connectivity status, and identify communication interruptions during UAV operation. In practical UAV communication environments, heartbeat messages are commonly utilized for system discovery, link monitoring, and real-time status synchronization between communicating nodes [13]. Since HEARTBEAT packets are transmitted periodically through wireless communication channels, they may become vulnerable to replay attacks, packet injection, or unauthorized manipulation in the absence of appropriate communication security mechanisms [1], [4].

Figure 2: illustrates the structure of the MAVLink HEARTBEAT message used for communication status monitoring between UAV nodes and the Ground Control Station. The message contains several operational fields including vehicle type, autopilot type, operating mode, system status, and MAVLink protocol version. These fields enable communication entities to monitor UAV activity, maintain connectivity awareness, and synchronize operational states during real-time UAV communication

type	autopilot	base_mode	custom_mode	system_status	mavlink_version
8 bits	8 bits	8 bits	32 bits	8 bits	8 bits

Figure 2:MAVLink HEARTBEAT Message Structure [13]

Table 2MAVLink HEARTBEAT Message Fields

Field	Size	Description
Type	8 bits	Specifies the UAV platform type
Autopilot	8 bits	Indicates the autopilot system used by the UAV
Base Mode	8 bits	Represents the UAV operating mode flags
Custom Mode	32 bits	Defines custom UAV operational modes
System Status	8 bits	Indicates the current UAV operational status
MAVLink Version	8 bits	Specifies the MAVLink protocol version

1.4 Security Requirements in UAV Communication Systems

Security in UAV communication systems is fundamentally defined through three core properties: confidentiality, integrity, and authentication. Confidentiality ensures that transmitted data remains accessible only to authorized entities by applying cryptographic encryption mechanisms. Integrity guarantees that data remains unaltered during transmission, ensuring that any modification can be detected using cryptographic hash functions or message authentication codes. Authentication ensures that communicating entities are legitimate and prevents unauthorized access or command injection. These properties collectively form the foundation of secure UAV communication design. Their interdependence is critical, as the absence of any single property significantly weakens the overall security posture of the system. Modern UAV security frameworks therefore rely on lightweight cryptographic mechanisms and authentication protocols specifically designed for resource-constrained environments [5], [6].

1.5 Cryptographic Mechanisms in UAV Systems

Cryptography plays a central role in securing UAV communication systems, particularly due to the limitations of wireless environments. Symmetric encryption algorithms are widely used because of their computational efficiency and suitability for real-time operations. In symmetric systems, a single shared secret key is used for both encryption and decryption, which ensures high performance but introduces challenges related to secure key distribution. Asymmetric cryptography, in contrast, uses a pair of mathematically related keys to facilitate secure key exchange and identity verification. While this approach provides stronger security guarantees, its computational complexity makes it less suitable for continuous telemetry encryption in UAV systems. As a result, hybrid cryptographic models are commonly adopted to balance efficiency and security requirements in UAV communication architectures [7], [10],[11].

1.6 ChaCha20 Encryption Algorithm

ChaCha20 is a modern symmetric stream cipher designed to provide high performance and strong security in software-based environments. It operates on a 512-bit internal state consisting of a 256-bit secret key, a 96-bit nonce, a 32-bit counter, and predefined constants. Through a sequence of quarter-round transformations involving modular addition, bitwise XOR, and rotation operations, ChaCha20 generates a pseudorandom keystream.

Encryption is performed by applying a bitwise XOR operation between the plaintext and the generated keystream, resulting in ciphertext. Decryption follows the same process due to the symmetric nature of the algorithm.

The security of ChaCha20 is based on the unpredictability of its keystream and resistance to known cryptanalytic attacks. Unlike traditional block ciphers that may require hardware acceleration for optimal performance, ChaCha20 is optimized for software execution, making it highly suitable for UAV systems with limited computational resources. Its consistent performance across different hardware platforms further enhances its applicability in real-time communication environments [7], [8],[9].

Equation (1): ChaCha20 Encryption is performed by applying a bitwise XOR operation between the plaintext message and the keystream generated using the ChaCha20 stream cipher. The generated keystream depends on the secret key, nonce value, and block counter in order to ensure secure and unique encryption for each transmitted packet.

$$\begin{aligned} & (1) \qquad C = P \oplus KS \\ & KS = ChaCha20(Key, Nonce, Counter) \end{aligned}$$

Where:

- *C*: Ciphertext
- *P*: Plaintext
- *KS*: Keystream generated by ChaCha20
- Key: secret encryption key
- Nonce: unique value used once
- Counter: block counter ensuring unique keystream blocks

Table 3: ChaCha20 Specifications

Parameter	Value	Description
Key Size	256 bits	Secret key
Nonce Size	96 bits (12 bytes)	Unique per packet
Block Size	512 bits	Internal structure
Operation	XOR	Encryption
Performance	High	Suitable for real-time

1.7 Poly1305 Authentication Algorithm

Poly1305 is a cryptographic message authentication code designed to ensure data integrity and authenticity. It operates by evaluating polynomial functions over a finite field defined by, producing a fixed-length authentication tag based on a secret key and the input message. Upon reception, the authentication tag is recomputed and compared with the transmitted value. Any mismatch indicates that the message has

been altered or corrupted during transmission. When combined with ChaCha20, Poly1305 forms an authenticated encryption scheme known as ChaCha20-Poly1305, which belongs to the class of Authenticated Encryption with Associated Data (AEAD). This construction provides confidentiality, integrity, and authentication simultaneously and is widely adopted in modern secure communication systems due to its efficiency and strong cryptographic guarantees [7], [8], [9].[12].

Equation (2): Poly1305 Authentication is used to generate a secure authentication tag for ensuring message integrity and sender authenticity. The authentication tag is generated using the secret key, ciphertext, and associated packet data before packet transmission.

$$(2) \quad TAG = Poly1305(K, C, A)$$

Where:

- *TAG*: Authentication tag
- *K*: Secret key
- *C*: Ciphertext
- *A*: Associated data (Header, SEQ)

Table 4: Poly1305 Specifications

Parameter	Value	Description
Key Size	256 bits	Authentication key
Tag Size	128 bits (16 bytes)	Integrity verification
Output	MAC Tag	Authentication
Security	High	Prevents tampering

The ChaCha20-Poly1305 authenticated encryption mechanism combines lightweight stream encryption with message authentication to provide confidentiality, integrity, and authenticity simultaneously. As illustrated in Figure 2, the encryption process begins by generating a pseudorandom keystream using the ChaCha20 algorithm based on a secret key and a unique nonce value. The generated

keystream is then combined with the plaintext message using the XOR operation to produce the ciphertext. In parallel, a subkey derived from the ChaCha20 keystream is utilized by the Poly1305 authentication algorithm to generate a 128-bit authentication tag. This authentication tag ensures message integrity and enables the receiver to detect any unauthorized modification or packet tampering during transmission. The integration of ChaCha20 and Poly1305 forms an efficient Authenticated Encryption with Associated Data (AEAD) scheme widely adopted in modern secure communication systems due to its strong security guarantees and low computational overhead.

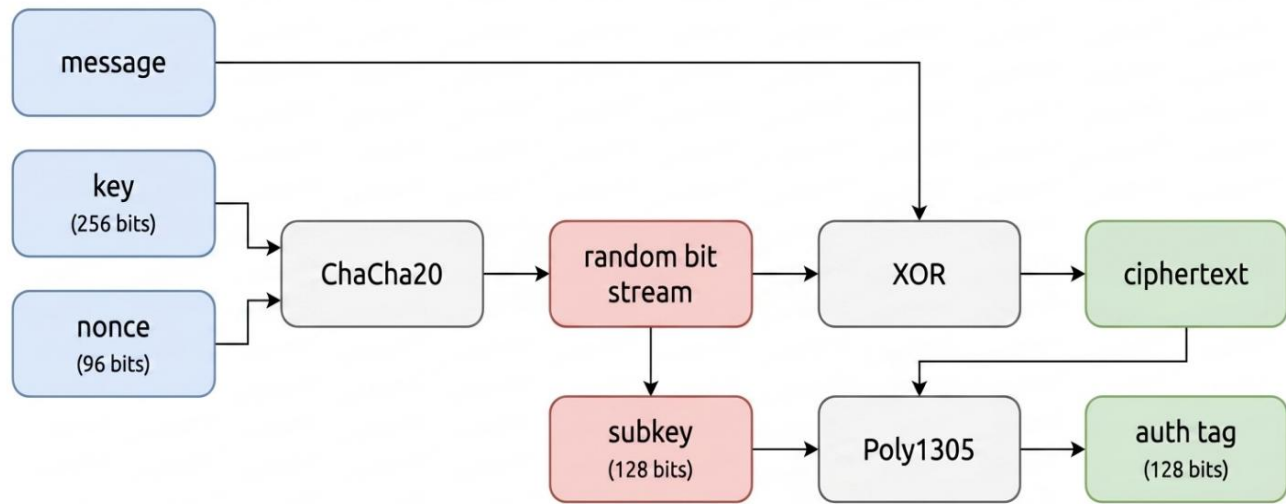


Figure3 :ChaCha20-Poly1305 Encryption [4]

Equation (3): AEAD Models generates encrypted ciphertext together with an authentication tag to provide confidentiality and integrity protection for MAVLink packet transmission.

$$(3) \quad (C, TAG) = AEAD_ChaCha20_Poly1305(K, Nonce, P, A)$$

The nonce (12 bytes) and authentication tag (16 bytes) are directly implemented using OpenSSL functions.

1.8 End-to-End Security in UAV Systems

End-to-end security refers to a communication paradigm in which data remains encrypted from the source UAV until it reaches the ground control station without being decrypted at intermediate nodes. This ensures that even if the communication channel is compromised, the attacker cannot access or manipulate the transmitted information.

In UAV systems, this model significantly reduces the attack surface by eliminating trust dependencies on intermediate communication infrastructure. It ensures that only the legitimate endpoints possess the cryptographic keys required to decrypt and interpret the data. Consequently, end-to-end encryption is considered a fundamental requirement for secure UAV communication in adversarial environments [5], [12].

1.9 Chapter Summary

This chapter presented the theoretical foundation of UAV communication systems with emphasis on system architecture, communication protocols, and security principles. The MAVLink protocol was analyzed as a lightweight communication standard optimized for efficiency but lacking intrinsic security mechanisms, making it vulnerable to various cyberattacks.

The fundamental security properties of confidentiality, integrity, and authentication were defined in the context of UAV systems. Furthermore, cryptographic mechanisms suitable for resource-constrained environments were discussed, highlighting the role of symmetric and asymmetric encryption approaches. ChaCha20 was introduced as a high-performance stream cipher optimized for software-based environments, while Poly1305 was presented as a robust authentication mechanism.

Finally, end-to-end security was described as a comprehensive model for ensuring full protection of UAV communication data.

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Chapter 2: Literature Review

2.1 Related Work

Recent years have witnessed significant research efforts aimed at enhancing the security of UAV communication systems, particularly those relying on the MAVLink protocol. Due to the lightweight nature and widespread adoption of MAVLink in platforms such as PX4 and ArduPilot, researchers have increasingly focused on developing secure communication frameworks capable of protecting telemetry and control messages against cyberattacks including replay attacks, spoofing, packet injection, and denial-of-service attacks.

Khan et al. proposed an additional security layer for MAVLink communication to improve the confidentiality and integrity of telemetry data exchanged between UAVs and Ground Control Stations (GCSs). Their framework introduced encryption mechanisms directly into MAVLink packet transmission while preserving low communication overhead.

Dixit et al. presented MAVShield, a lightweight cryptographic framework specifically designed for securing MAVLink-based UAV communication. The authors evaluated multiple encryption algorithms including AES-CTR, ChaCha20, Speck-CTR, and Rabbit cipher using a real UAV testbed environment.

Gómez Rey and Hernández Troncoso investigated the performance of AEAD-based encryption algorithms for securing MAVLink communication. Their study compared AES-GCM, ChaCha20-Poly1305, and TLS 1.3 using a Parrot Bebop UAV platform.

He et al. proposed a lightweight authentication and key-agreement protocol for UAV communication systems focusing on minimizing computational complexity while resisting replay and impersonation attacks.

Aljumah introduced a multilevel authentication framework for secure UAV communication within Internet of UAVs (IoUAV) environments using layered authentication and lightweight cryptographic operations.

Although these studies achieved considerable progress in UAV communication security, most existing approaches primarily focus on encryption and authentication while providing limited protection against replay attacks and duplicated packet execution.

Article Title	Publication Year	Journal / Source	Authors	What Was Achieved in This Paper	Reported Performance Metrics	Strengths	Relevance to our Project
Security Strategies for Unmanned Aircraft Systems Networks	2023	IEEE ICC Workshop	Marco Hernandez; Gurkan Gur; Kamesh Namuduri	Proposed a comprehensive security framework for UAS networks and analyzed threats such as spoofing, jamming, and DoS using a Zero-Trust approach.	Threat classification, vulnerability analysis of UAV communication layers	Comprehensive overview of UAV security challenges	Helps define the threat model for securing UAV–GCS communication
Security-Enhanced Lightweight Authentication Key-Agreement Protocol for UAV Communication	2025	Applied Sciences (MDPI)	Zhoucan He et al.	Introduced a lightweight authentication and key-agreement protocol suitable for resource-constrained UAV systems.	Computation cost, authentication delay, ECC operation count	Low computational overhead and resistance to replay/improsonation attacks	Supports designing lightweight authentication for UAV communication

Enhancing MAVLink Security in UAV Systems	2025	University Thesis / Research Study	Daniel Gomez Rey; Juan Pablo Hernandez Troncoso	Implemented encryption for MAVLink communication using AES-GCM, ChaCha20-Poly1305 and TLS 1.3.	Latency, RAM usage, communication overhead, battery consumption	Real UAV implementation and comparison of encryption methods	Directly relevant for securing MAVLink-based UAV communication
A Secure Communication Protocol for Unmanned Aerial Vehicle Networks	2022	Conference / Research Paper	Various Authors	Designed a secure protocol with encryption and mutual authentication to protect UAV network communication.	Authentication time, key generation cost, communication overhead	Improves confidentiality and resistance to network attacks	Provides protocol-level design ideas for secure UAV communication
UAV-Based Secure Data Communication: Multilevel Authentication	2024	Sensors (MDPI)	Abdullah Aljumah	Proposed a multi-level authentication model for Internet of UAVs	Latency, throughput, packet loss, energy consumption	Layered authentication with improved efficiency	Useful for designing multi-layer authentication

ation Perspective				using hashing, encryption and XOR operations .			mechanisms
A Novel Cipher for Enhancing MAVLink Security	2024	Journal / Conference in UAV Cybersecurity	Various Authors	Propose a new encryption algorithm to protect MAVLink messages and prevent data interception or modification	Encryption and decryption time, and the efficiency of the algorithm .	An algorithm tailored to the MAVLink protocol	Useful for developing a lightweight encryption layer for our project

2.2 Research Gap

Despite the extensive research conducted on securing UAV communication systems, several important limitations remain insufficiently addressed. Most existing approaches focus mainly on encryption mechanisms to provide confidentiality and integrity for MAVLink packets. However, encryption alone cannot prevent replay attacks in which an attacker captures valid packets and retransmits them at a later time to trigger unauthorized command execution. Many previous studies also rely heavily on simulation-based evaluations or theoretical analyses without implementing practical replay attack scenarios involving packet capture and retransmission. Another significant limitation concerns performance evaluation under lightweight UAV communication constraints. Some proposed solutions introduce noticeable computational, bandwidth, and latency overhead that may negatively affect real-time telemetry communication and flight stability.

Furthermore, limited research has explored modular secure communication architectures that can later be integrated into real UAV ecosystems such as PX4 autopilot systems and telemetry RF communication environments.

2.3 Research Contributions

This research proposes a lightweight secure communication framework for MAVLink-based UAV systems. The primary contributions of this work are summarized as follows:

- A. Designing a secure MAVLink communication architecture using the ChaCha20-Poly1305 authenticated encryption algorithm.
- B. Implementing an anti-replay protection mechanism based on sequence numbers and sliding-window validation.
- C. Simulating a replay attack scenario using an attacker node capable of retransmitting captured encrypted MAVLink packets.
- D. Measuring network performance metrics including latency, throughput, packet loss, and acknowledgment success rate.
- E. Developing a visual UAV communication environment using NetAnim and Wireshark integration.
- F. Designing the framework with modularity and extensibility to support future PX4 and telemetry RF integration.

2.4 Recent Advances in MAVLink and UAV Communication Security

Recent advances in UAV security research have increasingly focused on lightweight authenticated encryption algorithms optimized for embedded and resource-constrained systems. AEAD algorithms such as AES-GCM and ChaCha20-Poly1305 have become highly attractive due to their ability to simultaneously provide encryption and authentication with relatively low computational overhead.

ChaCha20-Poly1305 has received particular attention because of its efficiency in software-based implementations where hardware acceleration may not be available.

Another active research direction involves lightweight authentication and key-agreement protocols capable of securely establishing session keys under strict timing and computational constraints.

Researchers have also explored zero-trust architectures, secure telemetry routing, V2V authentication, and secure communication standards for Beyond Visual Line of Sight (BVLOS) UAV operations.

2.5 Challenges

Securing UAV communication systems presents several technical and operational challenges. UAV platforms are inherently resource constrained in terms of CPU capability, memory capacity, and battery power. UAV telemetry and control communication require extremely low latency and high reliability because flight commands and telemetry data are time-sensitive. MAVLink communication commonly operates over UDP or serial telemetry links, both of which lack built-in reliability and security guarantees. Replay attacks remain another major challenge because encryption alone cannot prevent attackers from retransmitting previously captured valid packets. Finally, real-world telemetry RF communication introduces practical constraints including limited bandwidth, high packet loss rates, interference, and communication instability.

2.6 Practical Applications

Secure MAVLink communication frameworks can be applied across numerous UAV domains requiring protected telemetry and command exchange. These include military reconnaissance, border surveillance, disaster response, smart agriculture, environmental monitoring, infrastructure inspection, emergency rescue operations, aerial mapping, and intelligent transportation systems.

The proposed secure communication architecture helps protect UAV control commands and telemetry data against eavesdropping, spoofing, replay attacks, unauthorized command injection, and packet manipulation attacks.

Furthermore, the modular design of the proposed framework enables future integration with PX4 autopilot systems, telemetry RF communication modules, and real UAV hardware platforms.

Chapter 3: Design and Implementation of a Secure MAVLink Communication System for UAVs

3.1 Introduction

This chapter presents the design and implementation of the proposed secure UAV communication framework. The system integrates MAVLink-based communication with a lightweight authenticated encryption mechanism using the ChaCha20-Poly1305 algorithm. The framework is implemented using the ns-3 simulator to provide secure and efficient communication between the Ground Control Station (GCS) and multiple UAV nodes operating in a wireless ad-hoc environment.

The proposed framework aims to ensure confidentiality, integrity, authentication, and replay attack protection while maintaining low communication overhead and real-time performance suitable for UAV applications.

3.2 Network Topology Design

The proposed communication system consists of one Ground Control Station (GCS) and two UAV nodes communicating through an IEEE 802.11g wireless ad-hoc network. An IPv4 addressing scheme based on the 192.168.1.0/24 network is utilized to support communication between all nodes.

The GCS is responsible for generating and transmitting encrypted MAVLink control packets toward the UAV nodes. In contrast, UAV nodes receive, authenticate, decrypt, and process incoming commands before execution. In addition, UAV nodes periodically transmit heartbeat and acknowledgment packets to maintain communication status awareness. The proposed architecture integrates MAVLink communication with an additional security layer based on ChaCha20-Poly1305 authenticated encryption. This integration enables packet confidentiality through encryption, integrity verification using authentication tags, and replay attack mitigation through sequence number validation.

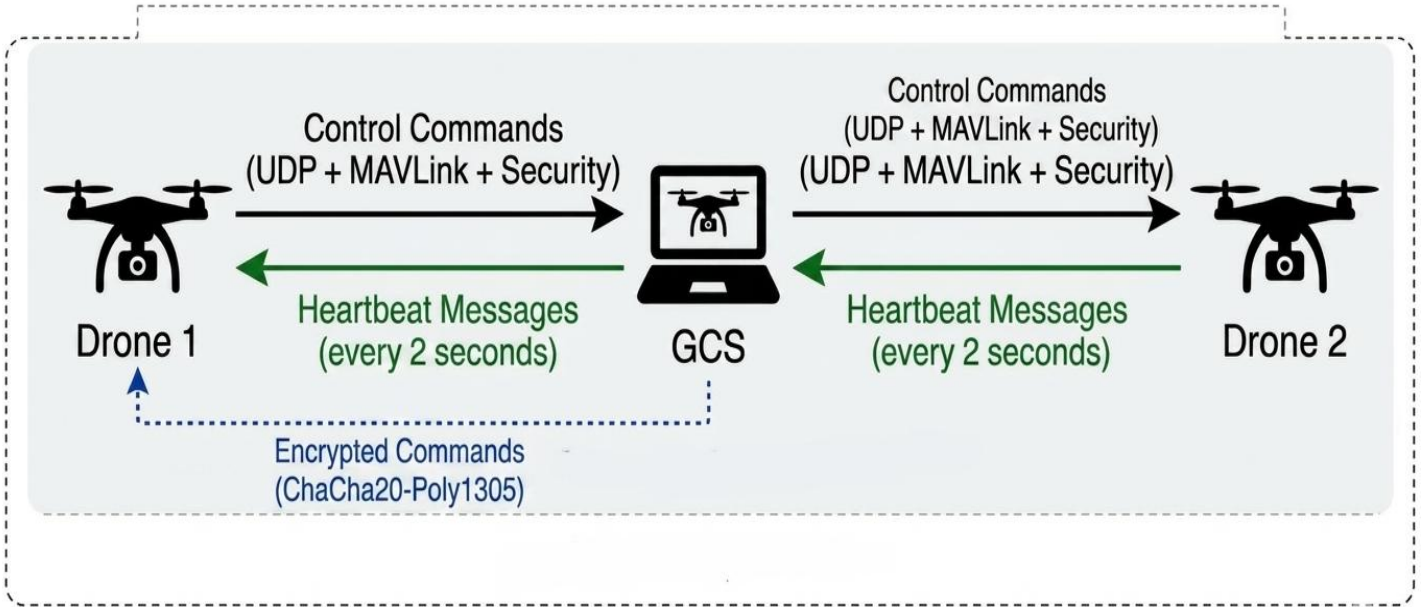


Figure4 : Network Topology Design

3.3 Implementation Parameters

The system configuration parameters are summarized as follows:

Table 5: Simulation Parameters for UAV Communication System

Parameter	Value	Description
Number of UAVs	2	Two UAV nodes
Simulation Time	12 s	Total simulation duration
Packet Size	32–64 bytes	Includes MAVLink + security overhead
Data Rate	1 Mbps	Wireless transmission rate
Transmission Interval	2 s	Packet sending interval
MAC Protocol	IEEE 802.11g	Ad-hoc wireless MAC
Transport Protocol	UDP	Connectionless transport protocol

3.4 MAVLink Message Types

The communication protocol supports a predefined set of control and status messages, which are exchanged between the GCS and UAV nodes.

These include:

- **CONN**: Used to establish a connection between UAV and GCS.
- **UP**: Command to increase altitude or move upward.
- **DOWN**: Command to decrease altitude or move downward.
- **STOP**: Command to halt UAV movement or operation.
- **HEARTBEAT**: Periodic message transmitted by UAVs to indicate operational status and connectivity.

These messages ensure continuous interaction and coordination between system entities.

The interaction between these message types and the secure communication workflow is illustrated in Figure 3. The figure demonstrates how encrypted MAVLink messages are exchanged securely between the Ground Control Station (GCS) and the UAV using the proposed communication framework. Each transmitted packet is protected using the ChaCha20-Poly1305 authenticated encryption algorithm in combination with sequence numbers, acknowledgment mechanisms, and anti-replay validation.

The communication process begins with a secure connection establishment message (CONN) transmitted from the GCS to the UAV using sequence number SEQ=1, followed by an encrypted acknowledgment (ACK) generated by the UAV. After successful connection establishment, the GCS sends encrypted control commands such as UP, DOWN, and STOP, each associated with incrementing sequence numbers to ensure packet freshness and replay attack prevention.

Meanwhile, the UAV periodically transmits encrypted HEARTBEAT and STATUS messages to maintain connectivity awareness and provide operational feedback to the GCS. In addition, each secure MAVLink packet consists of three primary components: Nonce, Ciphertext, and Authentication Tag. The Nonce ensures cryptographic uniqueness, the Ciphertext protects message confidentiality, and the Poly1305 authentication tag guarantees packet integrity and sender authenticity before packet acceptance.

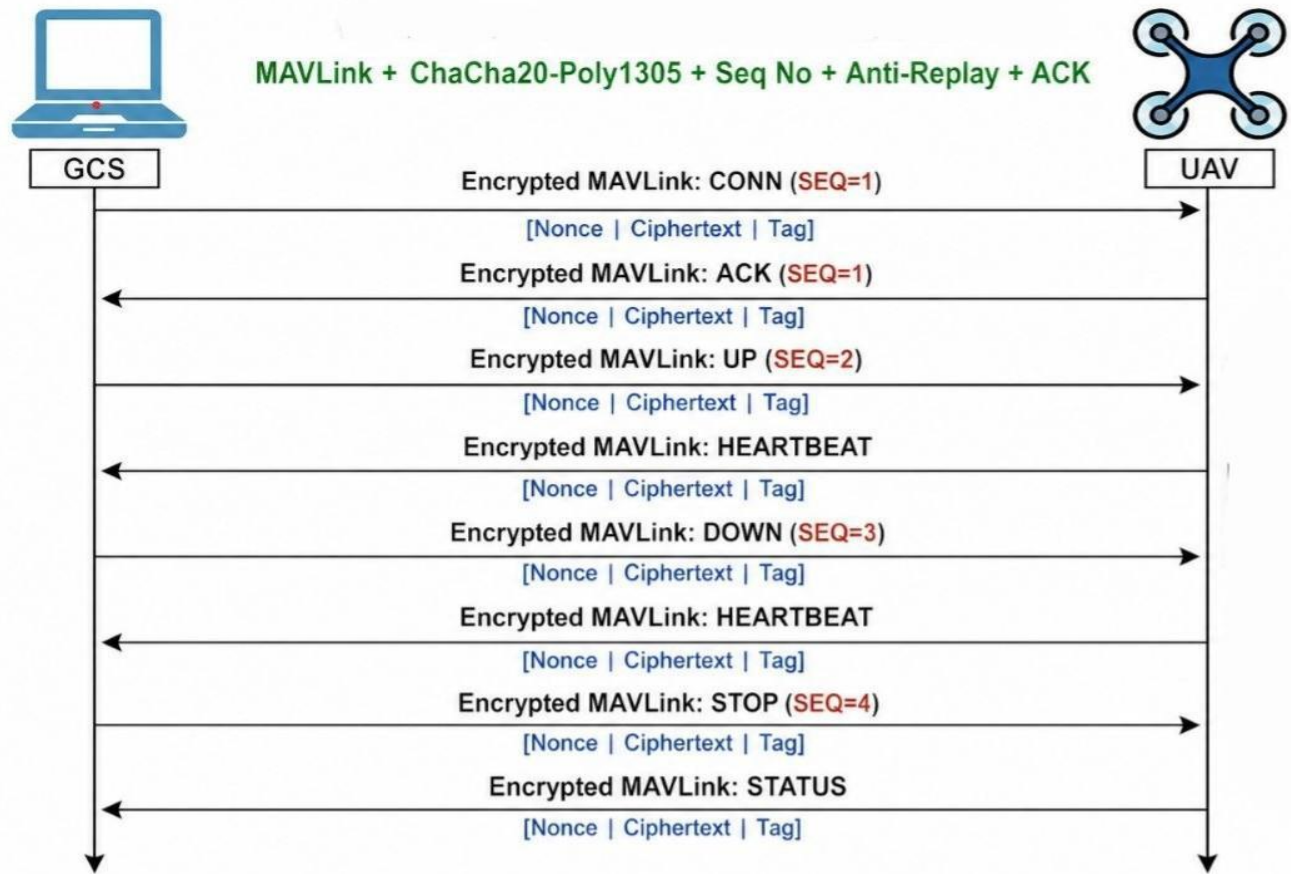


Figure5 : Secure MAVLink Communication Sequence Diagram

3.5 Secure Packet Structure

The proposed communication framework utilizes a secure packet structure designed to provide confidentiality, integrity, and authentication for MAVLink packet transmission between the Ground Control Station (GCS) and UAV nodes. Each transmitted packet is protected using the ChaCha20-Poly1305 authenticated encryption mechanism before wireless transmission.

The secure packet consists of three primary components: Nonce, Ciphertext, and Authentication Tag. The nonce is generated uniquely for each transmitted packet in order to ensure cryptographic uniqueness during the encryption process. The ciphertext contains the encrypted MAVLink payload generated using the ChaCha20 stream cipher, while the Poly1305 authentication tag is appended to provide packet

integrity verification and sender authentication before packet acceptance at the receiver side.

Table 6:Secure Packet Structure

Field	Size	Description
Nonce	12 Bytes	Unique encryption nonce
Ciphertext	Variable	Encrypted MAVLink payload
Authentication Tag	16 Bytes	Poly1305 authentication tag

The final secure packet format can be represented as follows:

Packet = Nonce || Ciphertext || Authentication Tag

This packet structure enables the proposed framework to provide lightweight secure communication suitable for real-time UAV communication environments

3.6 Sequence Number and Replay Protection

The Sequence Number (SEQ) field plays a fundamental role in maintaining both communication reliability and security within the proposed UAV communication framework. Each transmitted MAVLink packet is assigned a unique sequence number that increases incrementally for every new packet generated by the Ground Control Station (GCS) or UAV node.

At the receiver side, the sequence number is analyzed before packet acceptance in order to detect duplicated or previously transmitted packets. This mechanism enables the system to maintain proper message ordering and identify unauthorized packet retransmission attempts commonly associated with replay attacks.

Replay attacks occur when an attacker captures legitimate communication packets and retransmits them later to trigger repeated execution of valid commands. To mitigate this threat, the proposed framework integrates an anti-replay validation

mechanism based on sequence number tracking and freshness verification. Packets containing previously received sequence numbers are automatically rejected and discarded before command execution.

In addition, the sequence validation mechanism ensures that only fresh and valid packets are accepted by the receiving node, thereby preserving communication integrity, preventing duplicated command execution, and improving overall system reliability during UAV operation.

3.6.1 Replay Attack Workflow

A replay attack occurs when an attacker captures legitimate communication packets and retransmits them at a later time in an attempt to trigger repeated execution of valid UAV commands. In the proposed framework, replay attacks are mitigated using sequence number validation and a sliding-window anti-replay mechanism. During the attack scenario, the attacker node captures encrypted MAVLink packets transmitted between the Ground Control Station (GCS) and UAV nodes. The attacker then retransmits previously captured packets toward the UAV receiver without modifying the encrypted payload. Upon packet reception, the UAV first verifies the Poly1305 authentication tag to ensure packet integrity and authenticity. If authentication succeeds, the UAV validates the packet sequence number against previously accepted sequence values stored within the sliding-window replay protection mechanism. If the received sequence number has already been processed or falls outside the valid sequence window, the packet is classified as a replay attack and discarded immediately before command execution. Otherwise, the packet is accepted and processed normally. This mechanism enables the proposed framework to effectively detect and block replayed MAVLink packets while maintaining lightweight computational overhead suitable for real-time UAV communication environments.

3.7 System Architecture (OSI Model and Protocol Stack)

The proposed system follows a layered communication architecture based on the Open Systems Interconnection (OSI) model. Each layer is responsible for a specific communication functionality, allowing modular integration between networking, security, and application components.

The Application Layer implements a MAVLink-like communication protocol responsible for exchanging UAV control commands, telemetry information,

heartbeat messages, and acknowledgment packets between the Ground Control Station (GCS) and UAV nodes.

The Security Layer introduces authenticated encryption using the ChaCha20-Poly1305 AEAD algorithm. This layer provides confidentiality through ChaCha20 encryption while ensuring integrity and authentication using the Poly1305 authentication tag. Furthermore, replay attack mitigation is achieved through sequence number validation and freshness verification mechanisms.

At the Transport Layer, the User Datagram Protocol (UDP) is employed to support lightweight and low-latency communication suitable for real-time UAV telemetry and control transmission. UDP was selected due to its minimal overhead and suitability for time-sensitive communication environments.

The Network Layer utilizes the Internet Protocol (IP) to provide logical addressing and packet routing between network entities within the simulation environment.

At the MAC Layer, the system adopts IEEE 802.11 wireless communication mechanisms and the CSMA/CA medium access protocol to manage wireless channel access between communicating nodes.

Finally, the Physical Layer represents the wireless communication medium responsible for transmitting radio signals and electromagnetic waves between nodes through the simulated wireless channel environment provided by the ns-3 network simulator.

The layered architecture of the proposed UAV communication framework is illustrated in Figure 5. The figure demonstrates how MAVLink control and telemetry messages are processed across different communication layers, starting from the application layer and proceeding through the security, transport, network, MAC, and physical layers before wireless transmission occurs. In addition, the figure highlights the integration of the ChaCha20-Poly1305 security layer between the application and transport layers to provide encryption, authentication, and secure packet transmission capabilities.

3.8 Security Layer Design

The proposed framework integrates a lightweight authenticated encryption mechanism using the ChaCha20-Poly1305 AEAD algorithm. The security layer combines encryption, authentication, and replay protection within a unified communication framework. During transmission, the MAVLink payload and sequence number are combined and encrypted using the ChaCha20 stream cipher. After encryption, a Poly1305 authentication tag is generated and appended to the encrypted payload before packet transmission. At the receiver side, the packet undergoes multiple verification stages before command execution. Initially, the authentication tag is verified to ensure packet integrity and sender authenticity.

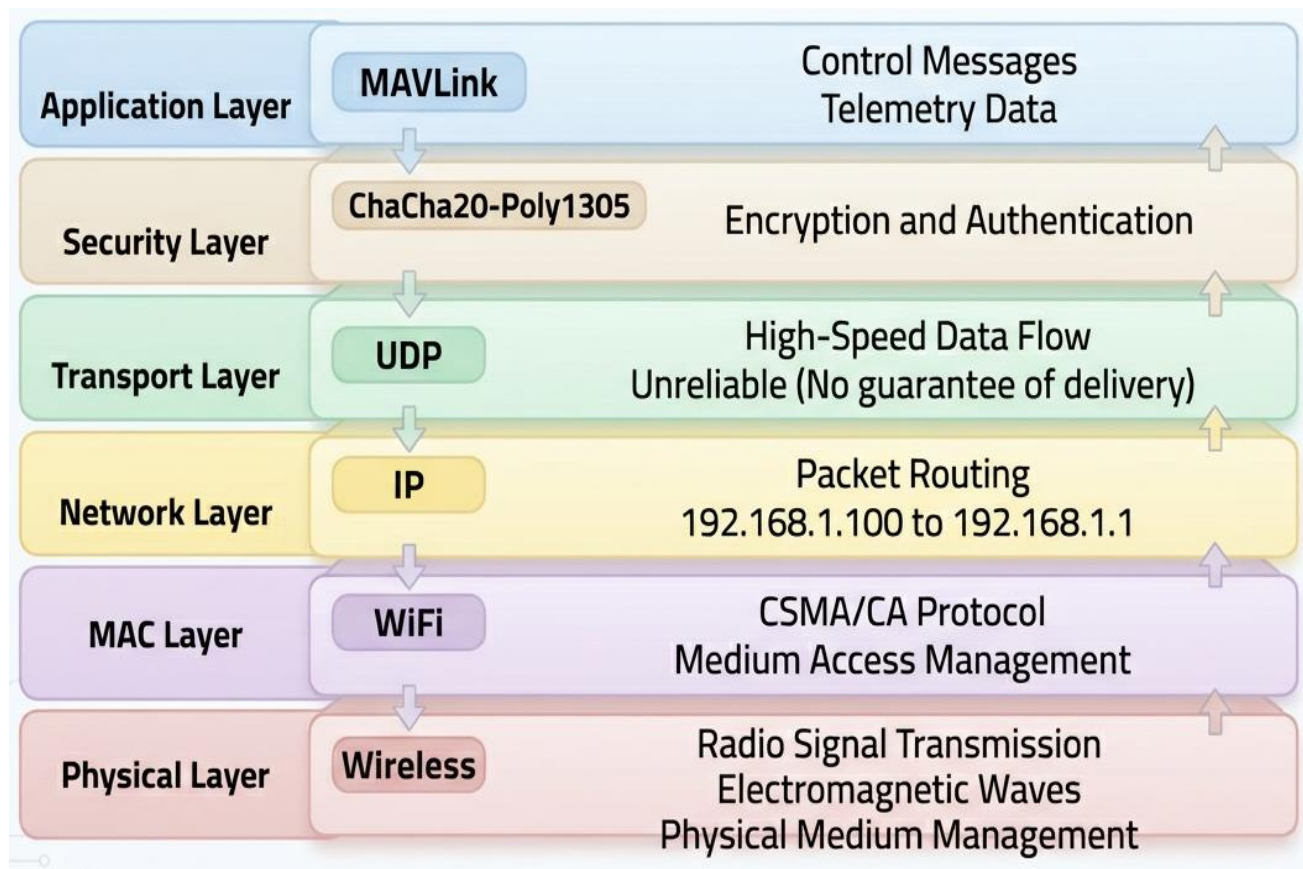


Figure6 : System Architecture (OSI Model and Protocol Stack)

Subsequently, the sequence number is validated to detect duplicated or replayed packets. Finally, the encrypted payload is decrypted and processed as a MAVLink message.

Packets failing authentication or replay validation are discarded immediately before UAV command execution. ChaCha20-Poly1305 was selected due to its computational efficiency, lightweight implementation, and suitability for resource-constrained UAV communication environments.

Figure 7 illustrates the secure communication architecture of the proposed framework.

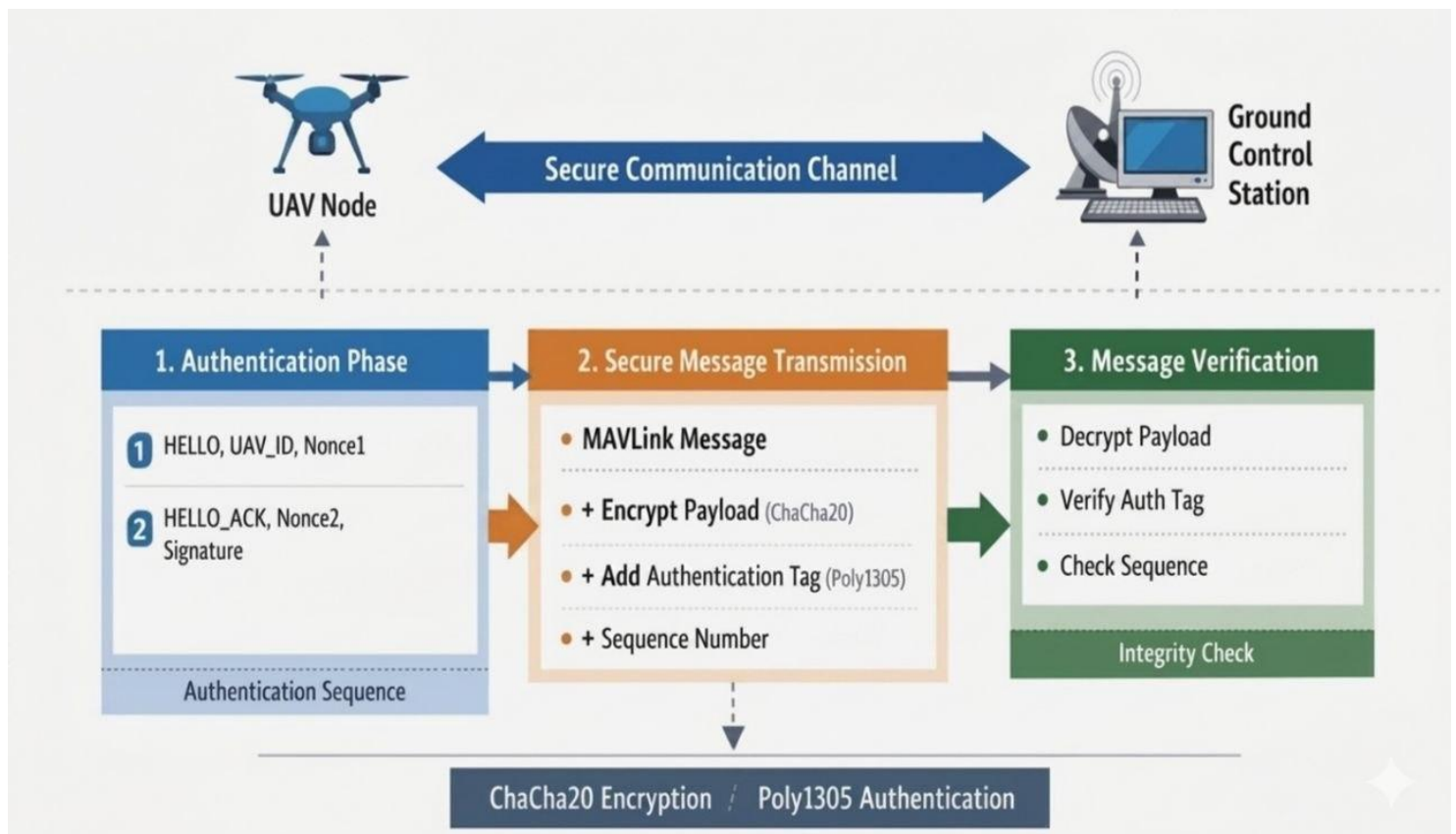


Figure7 : Secure UAV Communication Framework Based on ChaCha20-Poly1305

3.9 Secure Transmission and Reception Process

3.9.1 Transmission Process

The transmission process follows these steps:

- A. Create MAVLink-like message
- B. Add Sequence Number
- C. Encrypt payload using ChaCha20
- D. Generate authentication tag using Poly1305
- E. Transmit packet via UDP

3.9.2 Reception Process

The receiver performs the following steps:

Receive packet via UDP

- A. Verify authentication tag
- B. Detect replay attacks using SEQ
- C. Decrypt ciphertext using ChaCha20
- D. Parse MAVLink-like message

The overall workflow of the proposed secure MAVLink communication framework is illustrated in Figure 8. The figure describes the complete process of secure packet generation, transmission, verification, and replay attack prevention between the Ground Control Station (GCS) and the UAV node.

On the sender side, represented by the GCS, the communication process begins with generating a control command such as CONN, UP, DOWN, or STOP. Afterward, a sequence number (SEQ) and MAVLink payload are attached to the generated message in order to support packet freshness validation and replay attack protection.

Next, the payload is encrypted using the ChaCha20 stream cipher to ensure message confidentiality. Following encryption, an authentication tag is generated using the Poly1305 authentication mechanism to provide integrity verification and sender authentication. The final secure packet is then constructed in the form:

Packet = Nonce || Ciphertext || Authentication Tag

After packet construction, the secure packet is transmitted through the UDP communication channel toward the UAV node.

On the receiver side, represented by the UAV, the packet is first received via UDP and its size is validated to ensure structural correctness. The UAV then extracts the three primary packet components: Nonce, Ciphertext, and Authentication Tag. The encrypted payload is decrypted using ChaCha20, after which the Poly1305 authentication tag is verified.

The first decision stage evaluates whether the authentication tag is valid. If the authentication process fails, the packet is immediately discarded because the packet may have been modified, corrupted, or generated by an unauthorized source.

If the authentication tag is successfully verified, the system proceeds to the second validation stage, where the sequence number is checked. This stage determines whether the received sequence number is fresh and non-repeated. If the sequence number corresponds to an old or previously received packet, the packet is classified as a replay attack and discarded.

Finally, if both authentication and sequence validation succeed, the packet is accepted and the corresponding UAV command is executed safely.

The symbols used within the flowchart represent different operational components of the framework. Rectangular blocks indicate processing operations such as encryption, decryption, packet construction, and validation steps. Diamond-shaped symbols represent decision points where authentication or replay validation outcomes determine the subsequent execution path. Oval symbols indicate the start and termination points of the communication process, while directional arrows represent the sequence and flow of packet processing operations throughout the framework

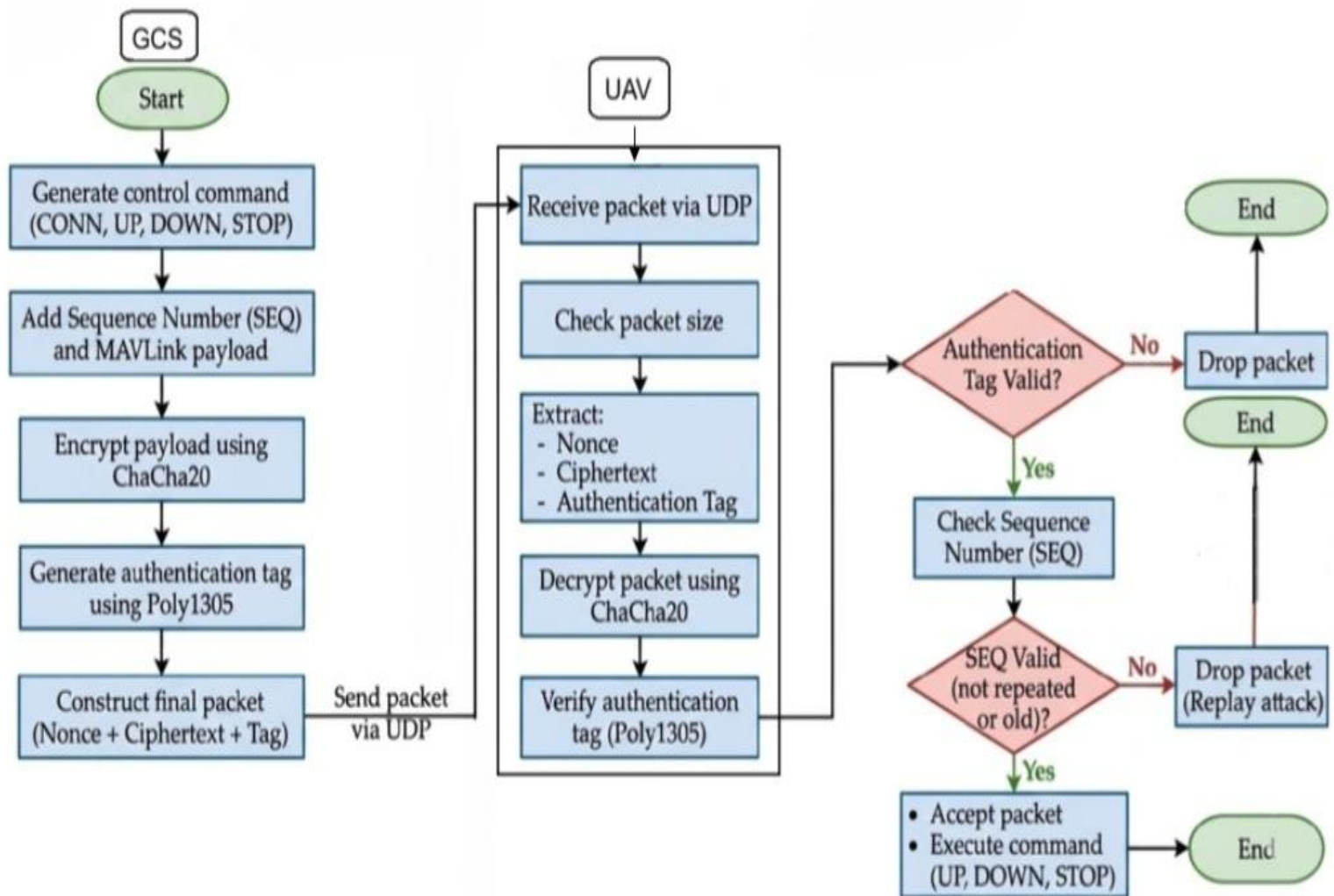


Figure 8: Secure Transmission and Reception Process

3.10 Project Implementation Phases

The overall implementation methodology of the proposed secure UAV communication framework is illustrated in Figure 9. The figure presents the sequential development and execution stages of the system, starting from simulation setup and ending with performance evaluation and visualization.

The process begins with the Simulation Setup stage using the ns-3 network simulator, which provides the simulation environment required for modeling UAV communication and network behavior. After initializing the simulation environment, the Network Configuration stage configures the wireless communication infrastructure using Wi-Fi ad-hoc networking and UDP communication protocols to enable lightweight peer-to-peer packet transmission between nodes.

Next, the MAVLink Implementation stage introduces MAVLink-based communication for transmitting UAV control commands and telemetry messages. Following that, the Packet Structure Design stage defines the internal structure of the transmitted secure packets, including MAVLink payloads, sequence numbers, cryptographic nonces, ciphertext, and authentication tags.

The framework then proceeds to the Security Integration stage, where the ChaCha20-Poly1305 authenticated encryption algorithm is integrated to provide confidentiality, integrity, and authentication for all transmitted messages.

Afterward, the Replay Protection stage introduces sequence number validation and anti-replay mechanisms to detect and reject duplicated or previously transmitted packets. This mechanism ensures communication freshness and prevents replay attack execution.

The Reception Process stage represents the UAV-side packet processing workflow, where incoming packets are received, decrypted, authenticated, and validated before command execution.

Following secure packet processing, the simulation enters the Simulation Execution stage, during which communication scenarios, attack simulations, and packet exchanges are performed within the ns-3 environment.

Subsequently, the Performance Evaluation stage analyzes network and security performance metrics such as latency, throughput, packet loss, replay detection efficiency, and communication reliability.

Finally, the Visualization stage utilizes NetAnim to provide graphical visualization of node movement, packet transmission, communication flow, and replay attack behavior within the simulated UAV network.

The symbols used within the flowchart represent different implementation stages of the proposed framework. Rectangular blocks indicate processing and implementation phases, while directional arrows represent the sequential workflow and dependency between system stages

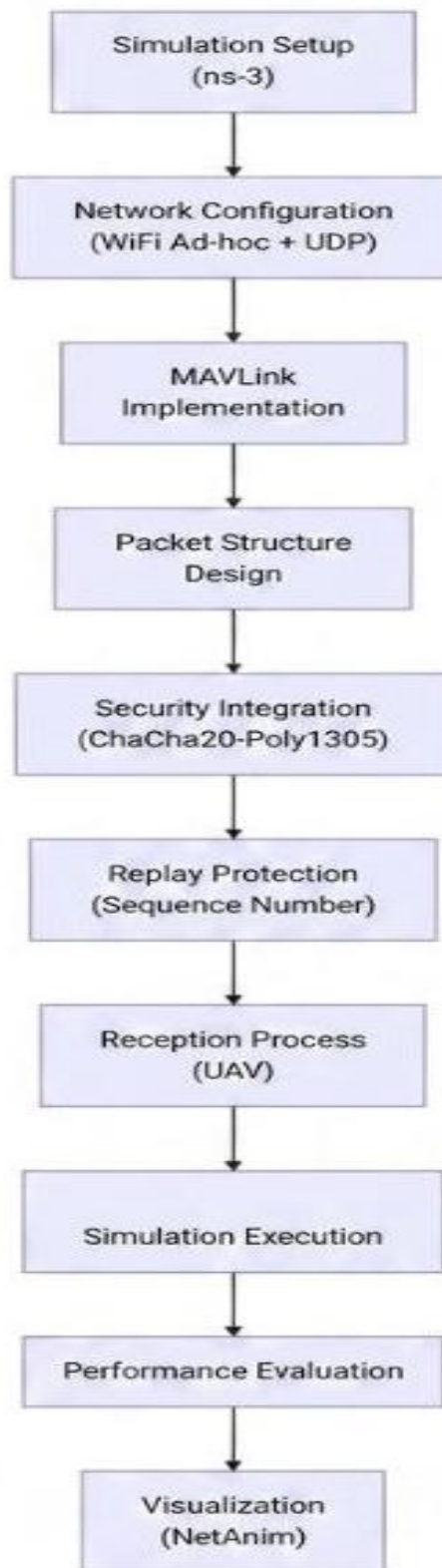


Figure9 : Sequential Implementation Process of the Proposed MAVLink Security Framework

3.11 Communication Scenario

The system supports bidirectional communication between the GCS and UAVs.

The GCS sends control commands, while UAVs transmit telemetry data such as GPS location, battery status, and system state.

3.12 Performance Metrics

The system evaluates performance using:

Equation (4): Packet Loss Rate is calculated by dividing the number of lost packets by the total number of transmitted packets during communication.

$$(4) \quad PLR = \frac{Sent - Received}{Sent} \times 100$$

Equation (5): Average Latency is calculated by dividing the total transmission delay of all received packets by the total number of successfully received packets.

$$(5) \quad Latency = \frac{\sum Delay}{Received}$$

Equation (6): Throughput is calculated by dividing the total number of successfully received bits by the overall simulation time.

$$(6) \quad Throughput = \frac{Total\ Bits}{Simulation\ Time}$$

Equation (7): Replay Acceptance Rate is calculated by dividing the number of blocked replay packets by the total number of replay attack attempts.

$$(7) \quad Replay = \frac{Accepted\ Replay\ Packets}{Total\ Replay\ Attempts}$$

3.13 Chapter Summary

This chapter presented the design and methodology of the proposed secure UAV communication system based on a MAVLink-like protocol integrated with ChaCha20-Poly1305 authenticated encryption. The system architecture was described using a layered model following the OSI reference stack, where secure communication is ensured at the security layer through lightweight AEAD encryption combined with sequence number tracking to mitigate replay attacks.

The cryptographic foundation of the system was detailed through the ChaCha20 stream cipher for encryption and the Poly1305 algorithm for message authentication, ensuring both confidentiality and integrity of transmitted data. Furthermore, the secure transmission and reception processes were outlined, demonstrating how UAV and GCS nodes handle encryption, authentication, and replay protection during communication.

In addition, the chapter defined the overall system architecture, communication scenario, and implementation phases, providing a complete understanding of how the proposed system operates. Finally, the performance evaluation metrics, including packet loss rate, average latency, throughput, and replay acceptance rate, were introduced to assess the efficiency and reliability of the system in later experimental analysis.

Overall, this chapter establishes the theoretical and structural foundation of the system, which is implemented and evaluated in the following chapter using the ns-3 simulation environment.

Chapter 4: Practical application and Results

4.1 Introduction

This chapter presents the practical implementation of the proposed secure UAV communication system. It describes the experimental environment, software tools, simulation setup, and testing procedures used to evaluate system performance. The chapter also highlights the implementation of the MAVLink-like protocol and the integrated security mechanisms in the ns-3 simulation environment.

4.2 Implementation Environment

The system is implemented using the ns-3 network simulator, which provides a discrete-event simulation platform for wireless networks. The implementation is written in C++ and integrates cryptographic libraries for security functions.

4.3 Software Tools and Libraries

The following tools and libraries are used in the implementation:

- ns-3 Network Simulator (v3.47)
- C++ Programming Language
- OpenSSL Library (for ChaCha20-Poly1305 encryption)
- MAVLink C Library
- Net Anim (for visualization)
- Wireshark (for packet analysis)

4.4 Hardware Environment (Simulation Setup)

Although the system is simulated, the following logical hardware setup is considered:

- Ground Control Station (GCS) Node
- UAV Node 1
- UAV Node 2
- Wireless Ad-hoc communication environment (IEEE 802.11b)

4.5 Simulation Configuration

The simulation parameters are defined as follows:

Table 7: UAV Network Simulation Configuration

Parameter	Value
Number of UAVs	2
Simulation Time	12 seconds
MAC Protocol	IEEE 802.11g Ad-hoc
Mobility Model	Constant Position
Transport Protocol	UDP
Channel Type	Wireless Channel

4.6 System Implementation Overview

The system implements a MAVLink-like protocol for communication between GCS and UAVs. The main implemented functions include:

- Command transmission (CONN, UP, DOWN, STOP)
- Heartbeat messaging for UAV status monitoring
- Replay attack simulation and detection
- Packet encryption using ChaCha20-Poly1305
- Sequence number tracking for integrity protection

4.7 Communication Interface

The communication is implemented using UDP sockets in ns-3. Each UAV node has a dedicated socket to receive commands from the GCS, while periodic heartbeat messages are sent back to maintain connectivity.

4.8 Visualization and Simulation Interface

Figure 10, At the beginning of the simulation, UAV nodes operate in an idle state to reduce energy consumption and avoid unnecessary processing. This design choice improves overall system efficiency, especially in resource-constrained UAV environments.

The following figure illustrates the initial simulation state in which the UAV nodes remain in idle mode to minimize unnecessary communication activity and reduce energy consumption until control commands are received from the Ground Control Station (GCS).

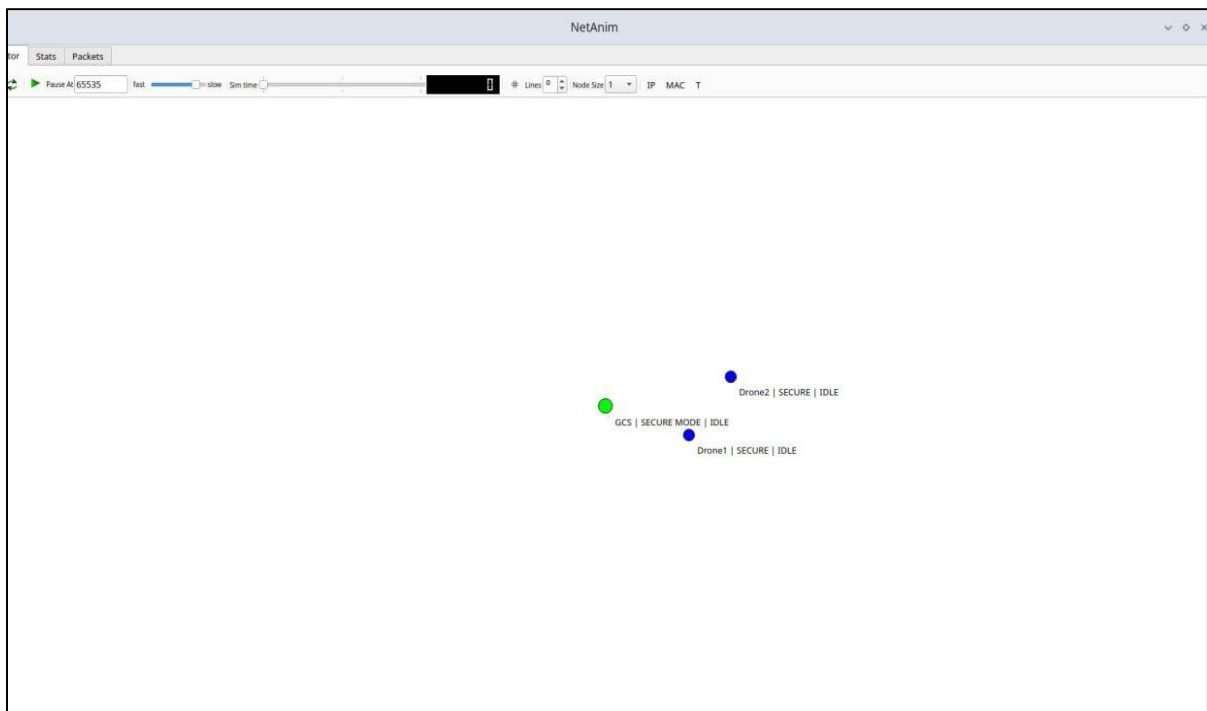


Figure10 Initial Simulation State Showing UAV Nodes in Idle Mode

Figure 11, The following figure illustrates the simultaneous transmission of packets from the Ground Control Station (GCS) to multiple UAV nodes during the simulation.

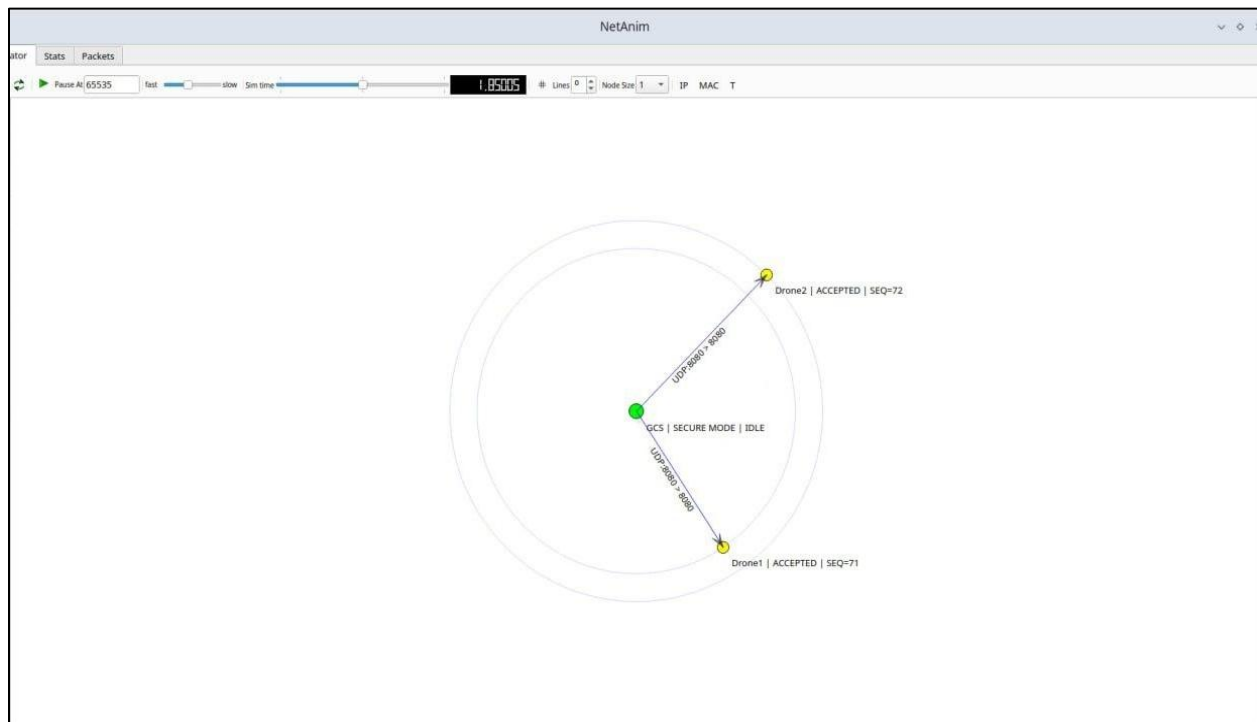


Figure 11: Simultaneous Packet Transmission from GCS to Multiple UAV Nodes

Figure 12, The following figure illustrates the reception of an acknowledgment (ACK) message by the Ground Control Station (GCS) from UAV 1, corresponding to a specific transmitted packet identified by its sequence number (SEQ).

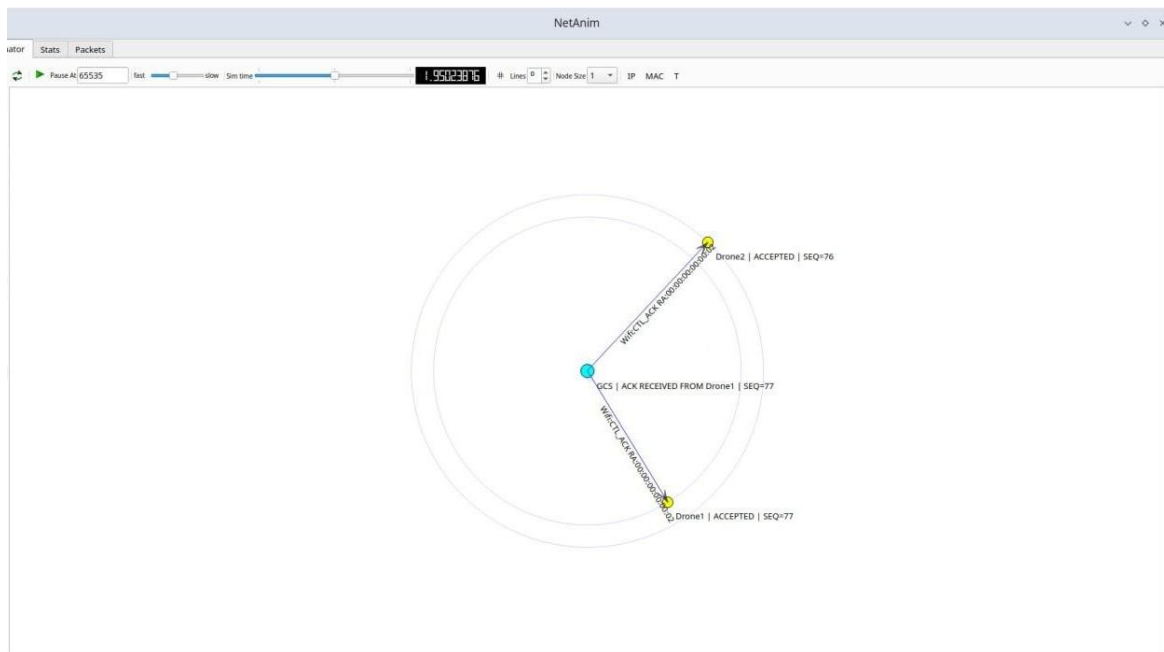


Figure 12:GCS Receiving ACK from UAV 1 with Corresponding Sequence Number (SEQ)

Figure 13, The following figure illustrates the addition of a fourth node representing an attacker in the network, which is used to simulate malicious activities such as jamming or packet injection.

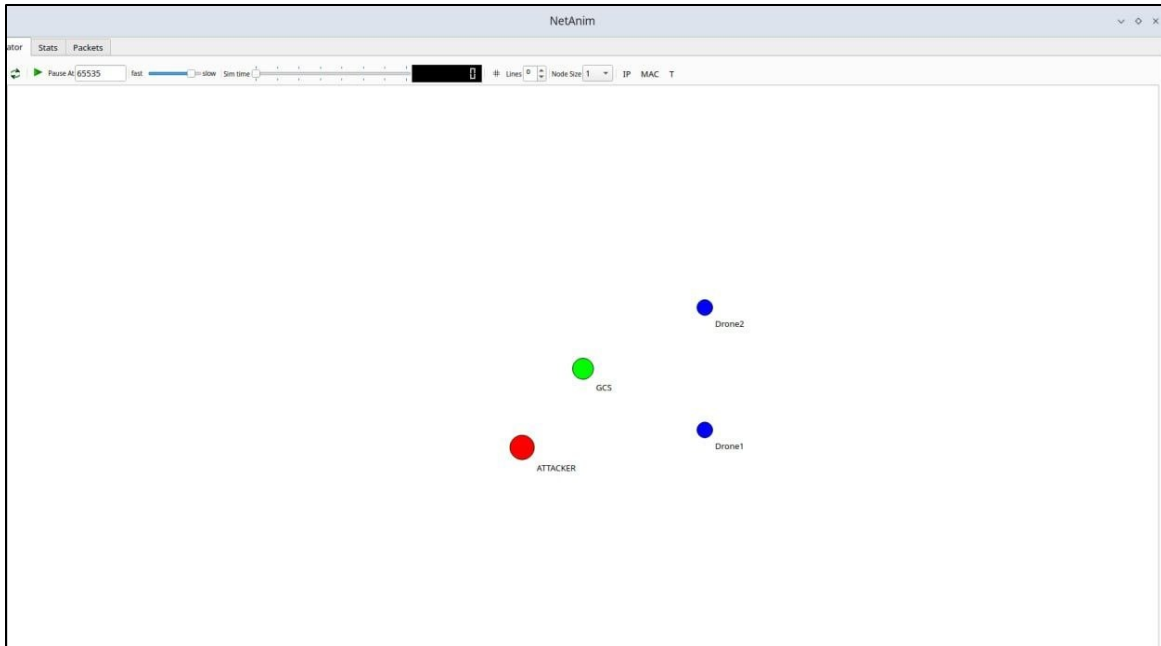


Figure 13: Network Topology with an Additional Attacker Node

Figure 14, The following figure illustrates the execution of a replay attack by the attacker node, where previously captured packets are retransmitted to the UAV nodes.

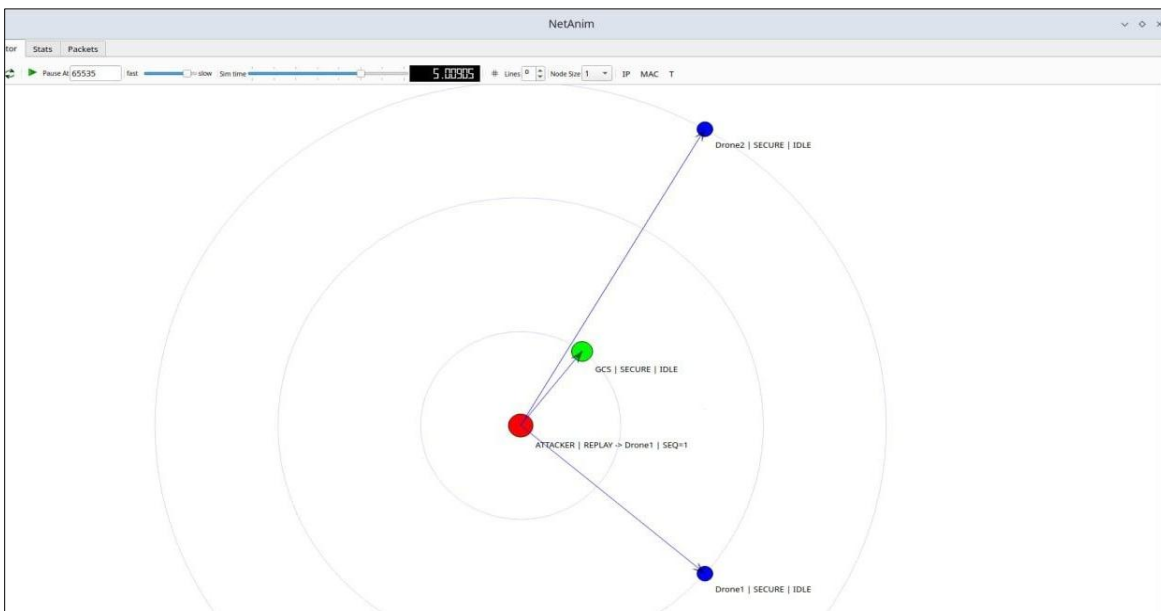


Figure14 : Replay Attack Executed by the Attacker Node on UAVs

Figure 15, 16 The following figure illustrates the detection of a replay attack by UAV (1) and UAV (2), where the node identifies the malicious packet based on sequence number validation and generates a warning message.

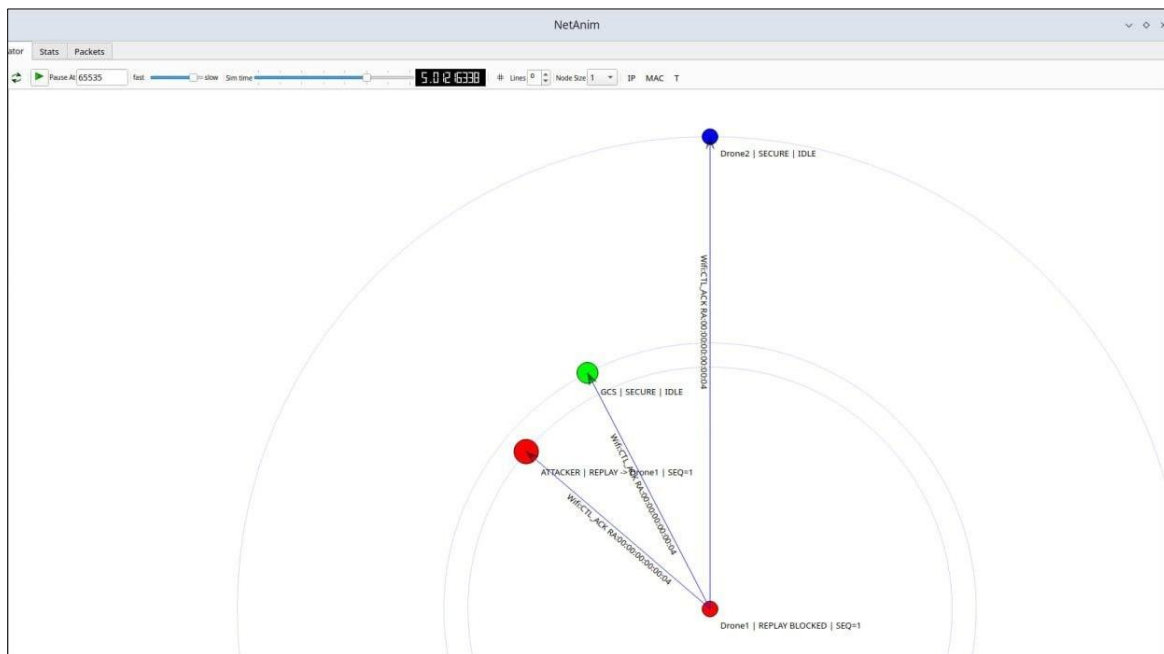


Figure15 Replay Attack Detection by UAV 1 and Warning Message Generation

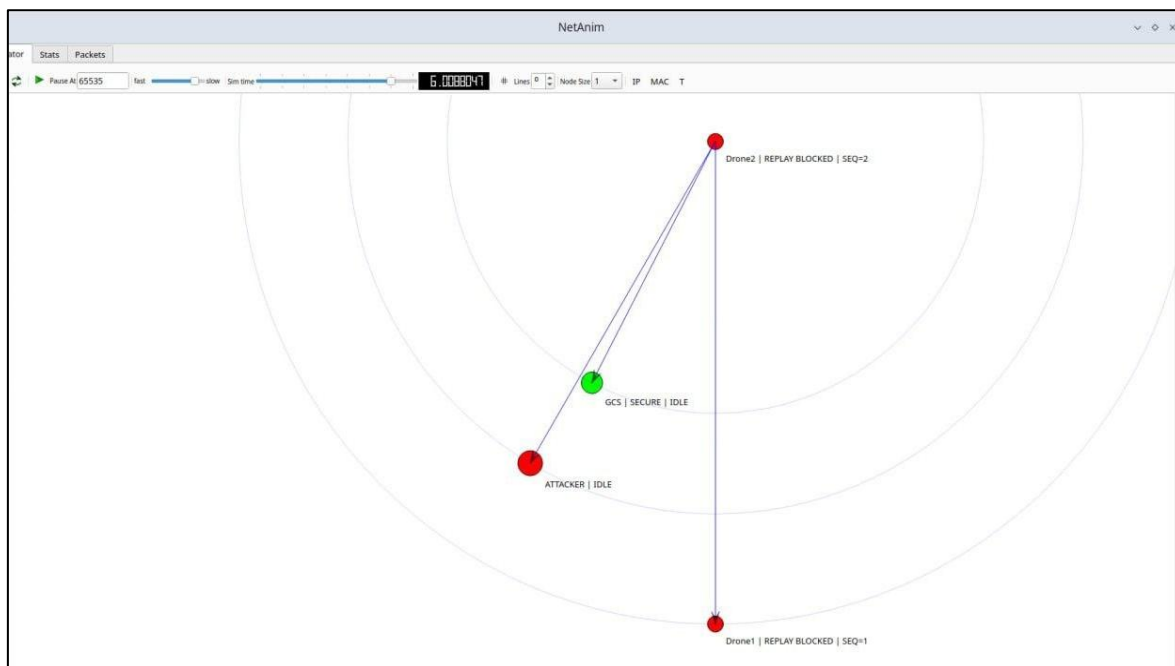


Figure16 :Replay Attack Detection by UAV 2 and Warning Message Generation

4.9 Testing Scenarios

The system is tested under different scenarios:

- Normal communication scenario (command exchange)
- Replay attack scenario
- Jamming / corrupted packet simulation
- Secure encryption/decryption validation

4.10 Performance Evaluation

The system is evaluated using the following metrics:

- Packet Loss Rate
- Average Latency
- Throughput
- Encryption/Decryption time
- Attack detection rate
- Energy efficiency

4.10.1 Results and Analysis

The following table compares the system performance before and after applying the security mechanism in terms of packet transmission and reliability.

Table 8: Performance Comparison Before and After Security Integration

Metric	BEFOR	AFTER
Original Sent Packets	100	100
Accepted Unique Packets	100	100
Packet Loss	0%	0%

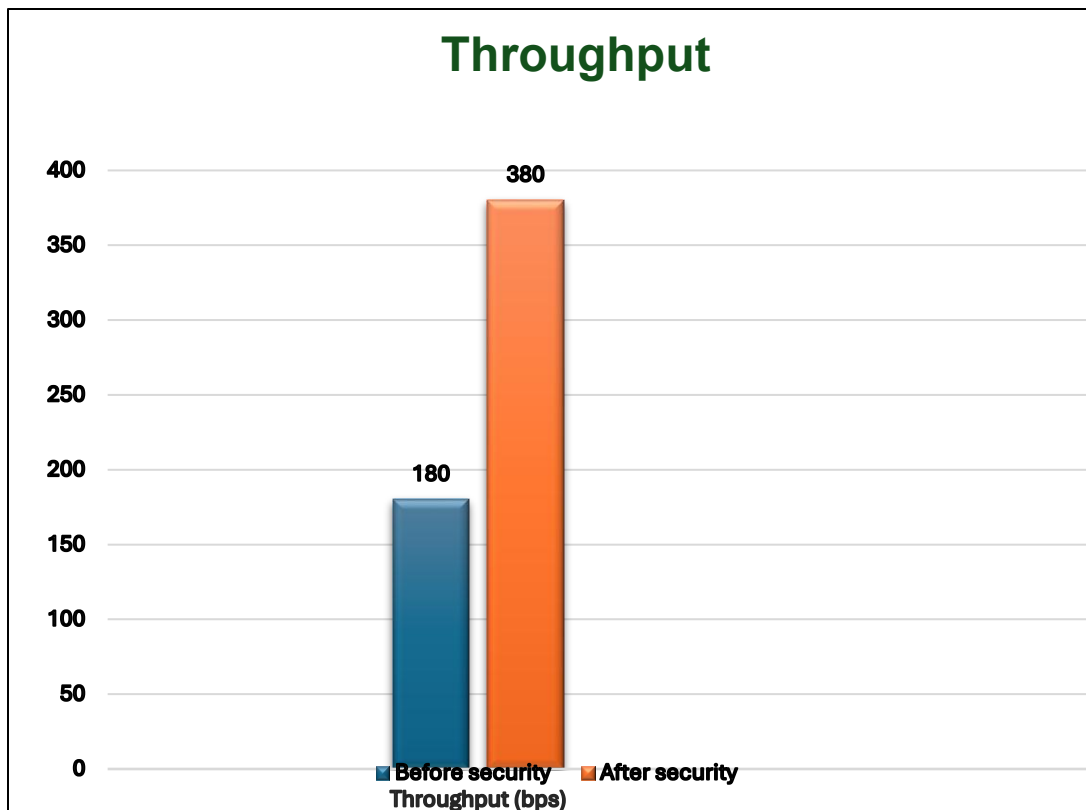


Figure17 : Throughput Comparison Before and After Security Implementation

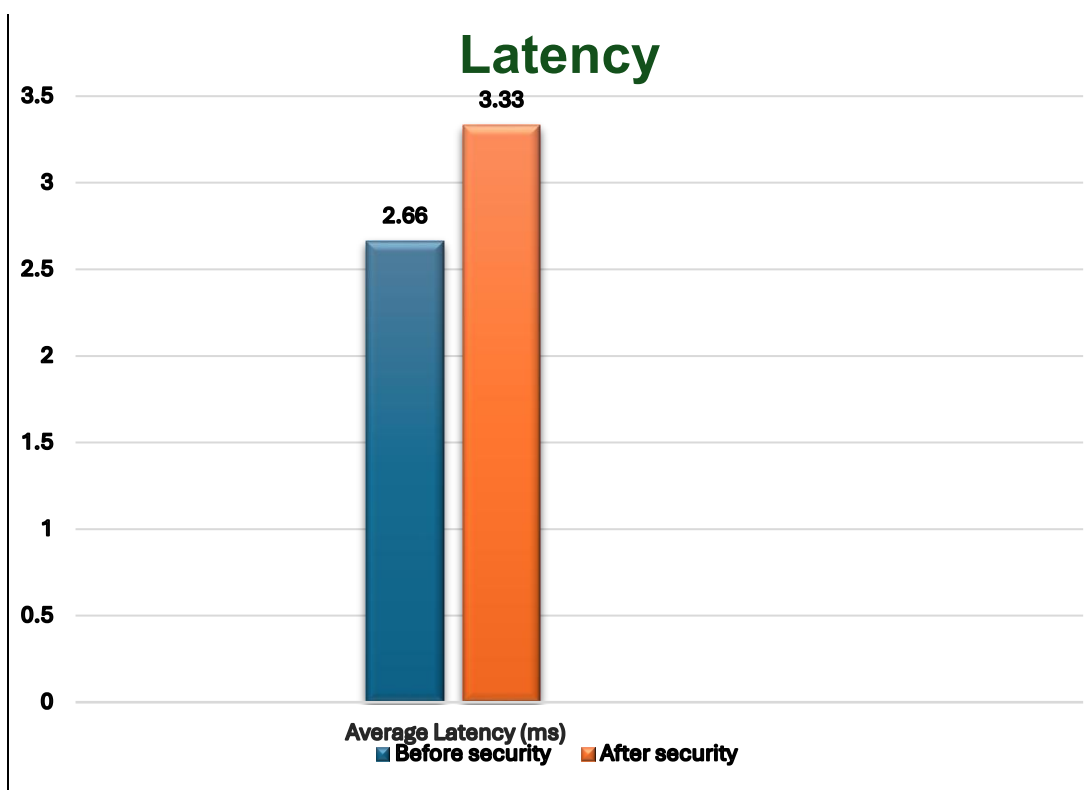


Figure18 Latency Comparison Before and After Security Implementation

The following table presents the security performance evaluation of the proposed system, including the number of authenticated packets and blocked malicious traffic.

Table 9: Evaluation of Security Performance Metrics

Metric	Value
Authenticated Packets	40
Blocked Packets	1948
Replay Attacks Blocked	1
Authentication Success Rate	98%

The following table summarizes the cryptographic performance of the ChaCha20-Poly1305 algorithm in terms of encryption and decryption time, as well as processing overhead.

Table 10: Cryptographic Performance of ChaCha20-Poly1305

Operation	Time
Encryption	91.6 μ s
Decryption	16.24 μ s
Security Overhead per Packet	28 bytes
Processing Impact	Low

The following table illustrates the additional overhead introduced by the security mechanism, including per-packet and overall network overhead.

Table 11: Security Overhead Analysis

Metric	Value	Description
Security Overhead per Packet	28 bytes	Added due to nonce + tag
Network Overhead	1400 bytes	Total additional transmission cost

The following table shows the impact of the security implementation on system energy consumption and overall efficiency.

Table 12: Energy Efficiency Evaluation

Metric	Value	Description
Final Energy	99.551 J	Remaining system energy
Efficiency	99.551%	Very low energy consumption impact

The following table provides a summary of attack detection results under different scenarios, including normal traffic, jamming, and replay attacks.

Table 13: Attack Detection Summary under Different Scenarios

Attack Type	Result	Count
Normal Traffic	Accepted	40
Jamming/Corrupted Packets	Blocked	1948
Replay Attack	Blocked	1

4.11 Chapter Summary

This chapter presented the practical implementation of the proposed UAV communication system. The ns-3 simulation environment successfully demonstrated secure and efficient communication between GCS and UAVs using a MAVLink-like protocol enhanced with ChaCha20-Poly1305 encryption.

Conclusion

This work presented the design and implementation of a secure UAV communication system based on a MAVLink-like protocol enhanced with ChaCha20-Poly1305 authenticated encryption. The primary objective was to address the security limitations of conventional MAVLink communication, particularly the lack of confidentiality, integrity, and authentication in UAV wireless networks.

A lightweight security layer was implemented at the application level, combining ChaCha20 encryption with Poly1305 authentication and sequence number tracking to mitigate replay attacks. The system was modeled using the OSI reference architecture and deployed over an IPv4-based IEEE 802.11g ad-hoc network using UDP for UAV–GCS communication.

The proposed system was evaluated using the ns-3 simulation environment under multiple scenarios, including normal operation and malicious attack conditions. The results demonstrated that the system effectively detects and blocks unauthorized and replayed packets while maintaining reliable communication performance.

Performance evaluation showed minimal overhead in terms of latency, throughput, and energy consumption, confirming the suitability of ChaCha20-Poly1305 for resource-constrained UAV environments. The cryptographic operations also exhibited low processing cost, ensuring real-time applicability.

Despite the promising results, the study is limited to simulation-based validation and simplified key management assumptions. Future improvements are required to extend the system beyond simulation environments.

In conclusion, the proposed system provides an efficient and lightweight security framework for MAVLink-based UAV communications, achieving a balanced trade-off between security strength and system performance for real-time aerial applications.

Future Work

Although the proposed system demonstrates strong performance and security in simulation, several enhancements can be considered for future development. First, integrating a dynamic key exchange mechanism would improve overall security by eliminating reliance on pre-shared keys. Second, extending the system to support large-scale UAV swarms would allow evaluation of scalability under dense network conditions.

Additionally, implementing and testing the proposed security framework on real UAV hardware would provide more realistic validation under physical channel conditions, including interference, mobility, and packet loss. Further optimization of energy consumption can also be explored to extend UAV operational lifetime during long missions.

Finally, future research may focus on enhancing attack detection capabilities by incorporating advanced intrusion detection mechanisms for UAV networks to improve resilience against sophisticated cyber threats.

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